

Syllabus

Spring Semester 2026

Course Title	Forensic Science	Course Code		KC6008	Credit	2	Classroom	Tamgu-gwan 2503
		Type		convergence	Hours	2		
Teacher	Name	Class No.	Office	Office Hour*	Contact		E-mail	
	Man-Seog Chun (천만석)	4	5602	Mon. 5, 6 Wed. 5 periods	+82-51-606-2340		manstone@ksa.kaist.ac.kr	
	Yeosan Lee (이여산)	2	2505	Tuesday 1-4 periods	+82-51-606-2223		yeosanksa@ksa.hs.kr	
	Wonjeon Kim (김원전)	1, 3	2103	Thursday 5, 6 periods			chem4u@naver.com	

* Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Description

Forensic Science is a two-credit introductory and convergent course that examines the foundational principles and methodologies of crime scene investigation (CSI) and forensic science. The course is designed to provide students with systematic training in core investigative procedures, evidence recognition and documentation, and the scientific reasoning applied in forensic contexts. The curriculum integrates concepts from physics, chemistry, anatomy, cell biology, environmental science, and computer science, emphasizing their application to forensic problem-solving. The course activities include lectures, structured laboratory exercises, guided case-study analyses, and online learning modules. Upon completion of the course, students will be expected to demonstrate an understanding of fundamental forensic concepts and to apply evidence-based scientific approaches to the analysis and interpretation of investigative questions.

2. Core Target Competencies

Subject Core Competencies						
analytical and creative thinking	problem- solving ability	science research ability	digital and quantitative literacy	leadership and teamwork	complex communication	adaptability /progression /spirit of adventure
* Blank fields to be filled in accordingly.						
* Fill in the blanks with descriptive words.						

3. Textbooks and References

3.1. Textbook

- (1) Textbook: Experiment protocol and materials provided by instructors

3.2. References

- (1) Criminalistics: An Introduction to Forensic Science, by Richard Saferstein
 (2) http://wps.prenhall.com/chet_saferstein_criminalistics_9/

- (3) Korean Supreme Prosecutors Office homepage <http://www.spo.go.kr/spo/index.jsp>
- (4) Crime lab, FBI <http://www.fbi.gov/homepage.htm>
- (5) Counterfeiting and Forgery <http://www.moneyfactory.com/>

4. Class Method

4.1 Class Management

- (1) The course is conducted through student-centered learning activities. Students will work in teams of two or three to complete course activities together.
- (2) Students work in teams and participate in team-based problem-solving activities during class. Through structured discussions, they examine the learning objectives and, by collaboratively solving assigned problems, develop a solid understanding of key scientific concepts and related knowledge.
- (3) Students autonomously form small groups to discuss the learning objectives. They then collaborate in small-group activities to solve problems and present their understanding to their peers. Finally, they share the course materials developed as a team and write reflective essays based on what they have learned.
- (4) Students are expected to cultivate self-directed and independent learning competencies. Collaboration and communication demonstrated through team-based projects will be evaluated in detail.
- (5) Reports, project outcomes, and other course materials must be submitted by the stated deadline using the designated submission method. Late submissions, non-submissions, or submissions made through unauthorized means will result in significant penalties in the evaluation.
- (6) After each experiment, clean and tidy up the experiment equipment and place used.

4.2 Academic Integrity Policy

- (1) The diversity of the class members ought to be respected. Any type of discrimination regarding ethnicity, nationality, language, personal orientation, etc., will not be tolerated.
- (2) Any type of misconduct (e.g., cheating, copying, possession of electronic devices during the exam, refusing to follow instructions, dishonorable actions, other unauthorized activities not related to the course, etc.) will automatically result in an F grade for the course. Furthermore, the student will receive official disciplinary punishment according to the Korea Science Academy Regulations.
- (3) Plagiarism and copying are strictly prohibited. Citations and references must be provided on a sentence-by-sentence basis whenever a source is used. Any sequence of six or more consecutive words taken from another source will be regarded as plagiarism. Note that paraphrasing is required whenever sources are referred to. All visual and technical materials (e.g., figures, equations, graphs, tables, and code) must be properly cited, and any reuse must comply with applicable permissions and licensing. Students must comply with all other regulations regarding academic integrity.
- (4) Using AI tools (e.g., ChatGPT) for academic support is permitted; however, students must verify and study the original sources cited or referenced. In addition, students must cross-check information using Google and multiple scholarly references. In this course, AI-generated content is not accepted as a citable source or reference. Submitting AI-generated content without proper citation to original sources will be regarded as plagiarism and unauthorized copying.

5. Evaluation

5.1 Evaluation Criteria

Pre-lab Report	Lab Activity and Report	Projects	Mid-term Exam	Final Exam	Attendance and Attitude	Sum
12%	36%	16%	12%	14%	10%	100%

5.2 Evaluation Method

5.2.1 Pre-lab Report (12%)

- (1) A pre-lab report that includes background information relevant to the experiment must be submitted before class begins. The topics are available in the [7. Weekly Plans] section below.
- (2) The pre-lab reports must be handwritten on A4-sized paper (1–2 pages).
- (3) The pre-lab reports must be submitted by the stated deadline using the designated submission method. Late submissions, non-submissions, or submissions made through unauthorized means will result in significant penalties in the evaluation.

5.2.2 Lab Activity and Report (36%)

- (1) Laboratory activities will be conducted during class. The points assigned to this criterion are 20%.
- (2) Laboratory worksheets (reports) will be provided for each activity. Each worksheet must be completed by hand and submitted before leaving class. The points assigned to this criterion are 16%.
- (3) Excused absences and medical absences guarantee attendance points in accordance with the Korea Science Academy Regulations. However, penalties for not conducting the lab activity in the designated class hours and for not submitting the corresponding worksheet still apply.
- (4) The worksheets must be submitted by the stated deadline using the designated submission method. Late submissions, non-submissions, or submissions made through unauthorized means will result in significant penalties in the evaluation.

5.2.3 Projects (16%)

- (1) The Projects criterion evaluates (a) presentations on forensic case studies and (b) a team-based forensic performance project.
- (2) The (a) presentations on forensic case studies focus on scientific methods used to resolve crime cases. The points assigned to this criterion are 6%.
- (3) In the (b) team project, class members will be divided into two teams. Each team will construct a simulated crime scene in the forensic room based on a forensic scenario. The other team will conduct forensic investigations to resolve the simulated case. The points assigned to this criterion are 10%.
- (4) The project scenario and final report must be submitted by the stated deadline using the designated submission method. Late submissions, non-submissions, or submissions made through unauthorized means will result in significant penalties in the evaluation.

5.2.4 Mid-term (12%) and Final (14%) Exams

- (1) The course is scheduled to include a midterm exam and a final exam.
- (2) The exams may include fill-in-the-blank, short-answer, and/or multiple-choice questions, among others.
- (3) Along with the other criteria, the exams will be administered in strict accordance with the Korea Science Academy Regulations and the [4.2 Academic Integrity Policy] section.
- (4) The make-up score for an exam missed due to an excused absence or a medical absence shall be calculated as follows: determine the student's percentile rank based on the total score from other exams in the same course that the student has taken, assign the score on the missed exam corresponding to the same percentile, and then multiply it by 100% (excused absence) or 80% (medical absence). Further details are available in the Korea Science Academy Regulations.

5.2.5 Attendance and Attitude (10%)

- (1) A 0.5% penalty will be applied for each late arrival to class.

- (2) A 1% penalty will be applied for the first unexcused absence. The second unexcused absence will result in an additional 3% penalty, and the third in a 5% more penalty. (Total 9% of penalty for three unexcused absence). Four or more unexcused absences will result in a FAIL grade for the course.
- (3) Assessment under the Attitude criterion will be subject to the instructor's discretion.
- (4) Sleeping, disrupting the class, disobeying instructions, etc., may result in significant penalties.
- (5) A 1% penalty will be applied for each violation of the tidiness regulations.

6. Criteria for Awarding Credits

- (1) The course will follow an absolute grading (criterion-referenced grading) system. Grades will be assigned based on the over evaluation points that each student achieves.
- (2) The grading scale is as follow: 100-90: A+, 90-85: A0, 85-80: A-, 80-75: B+, 75-70: B0, 70-65: B-, 65-60: C+, 60-55: C0, 55-50: C-, 50-45: D+, 45-40: D0, 40-35: D-, 35-: F.
- (3) Students with outstanding academic achievement may be recommended for the Academic Excellence Award and the Principal's Award, and their performance is recorded under the core competencies in the Subject Details. However, students with poor academic attitude or poor attendance will be excluded from consideration.

7. Weekly Plans

Week (period)	Topic and Contents	Notes
Week 1 2.23.~3.1.	Introduction Overview of Forensic Science Preparation for the document forgery activity	Twelve categories for document forgery
Core Competencies	Digital and quantitative literacy, leadership and teamwork, complex communication	
Week 2 3.2.~3.8.	Document Forgery	What is an innocent project?
Core Competencies	analytical and creative thinking, science research ability, problem-solving ability, digital and quantitative literacy, leadership and teamwork, complex communication	CSI effect paper survey 3.2. Substitute Holiday
Week 3 3.9.~3.15.	Presentation (CSI effect or innocence project)	Fingerprint pattern
Core Competencies	analytical and creative thinking, digital and quantitative literacy, leadership and teamwork, complex communication, adaptability/progression/spirit of adventure	
Week 4 3.16.~3.22.	Fingerprint	Suzuki and Tsuchihashi's classification
Core Competencies	analytical and creative thinking, digital and quantitative literacy, problem-solving ability, leadership and teamwork, complex communication	
Week 5 3.23.~3.29.	Lip print	Thin-layer chromatography with paper or silica

Core Competencies	analytical and creative thinking, digital and quantitative literacy, problem-solving ability, leadership and teamwork, complex communication	Composition of ink, lipstick, or drugs
Week 6 3.30.~4.5.	TLC of (ink or drug)	Preparation for the case study presentation
Core Competencies	analytical and creative thinking, digital and quantitative literacy, problem-solving ability, leadership and teamwork, complex communication	
Week 7 4.6.~4.12.	Presentation (case study)	Microscope operation Hair analysis
Core Competencies	analytical and creative thinking, digital and quantitative literacy, leadership and teamwork, complex communication, adaptability/progression/spirit of adventure	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Analysis of Hair	Blood spatter analysis
Core Competencies	analytical and creative thinking, digital and quantitative literacy, problem-solving ability, leadership and teamwork, complex communication	
Week 10 4.27.~5.3.	Prepare simulated blood and pattern analysis of blood spatter	Blood detection
Core Competencies	analytical and creative thinking, digital and quantitative literacy, problem-solving ability, leadership and teamwork, complex communication	
Week 11 5.4.~5.10.	Blood detection	Reports on specific fields in forensic science 5.5. Children's Day
Core Competencies	analytical and creative thinking, digital and quantitative literacy, problem-solving ability, leadership and teamwork, complex communication	
Week 12 5.11.~5.17.	Presentation (I am an expert in forensic science)	Casting methods
Core Competencies	digital and quantitative literacy, complex communication, leadership and teamwork, science research ability, analytical and creative thinking, adaptability/progression/ spirit of adventure	
Week 13 5.18.~5.24.	Casting	Preparation for the team project
Core Competencies	problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, adaptability/progression/ spirit of adventure	

Week 14 5.25.~5.31.	Team project : forensic performance	Preparation for the team project 5.25. Substitute Holiday
Core Competencies	problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, adaptability/progression/spirit of adventure, analytical and creative thinking	
Week 15 6.1.~6.7.	Team project : forensic performance	Preparation for the team project reports 6.3. Local Election Day
Core Competencies	problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, adaptability/progression/spirit of adventure, analytical and creative thinking	
Week 16 6.8.~6.14.	Make-up class week	-
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	-

8. Note

The course plan and policies may change at the instructor's consideration, depending on the given circumstances.